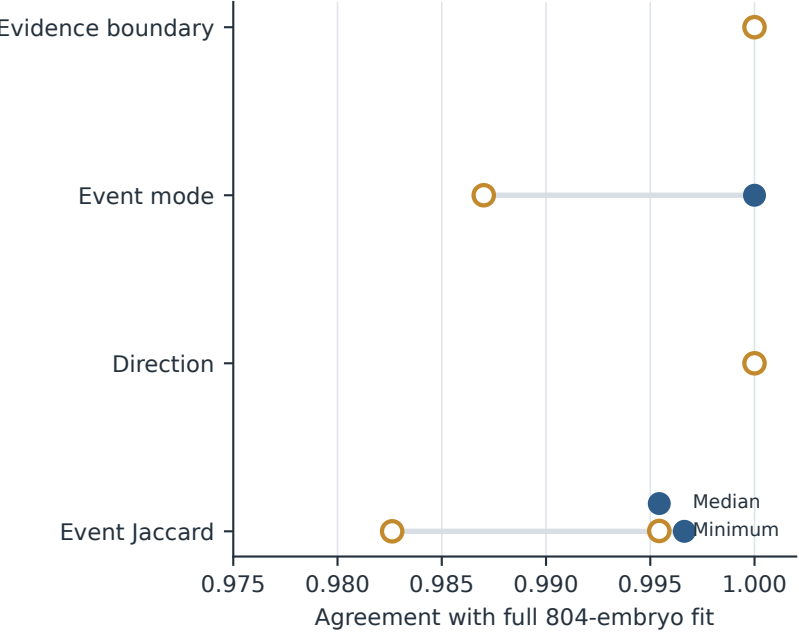
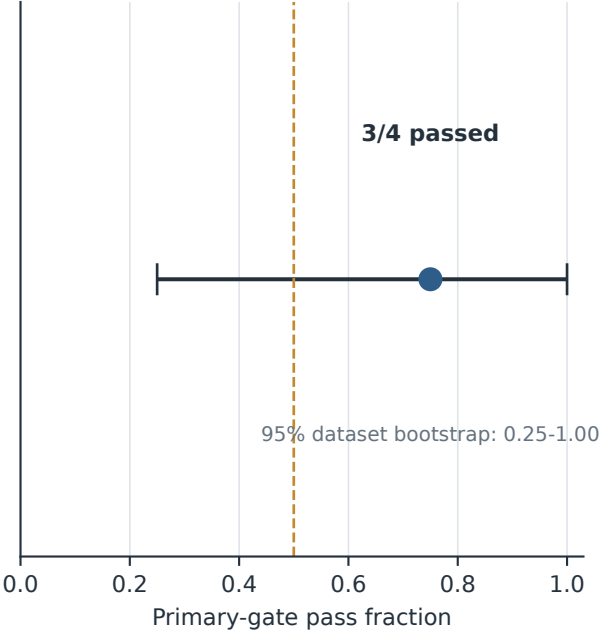


Robustness, portability and scalability

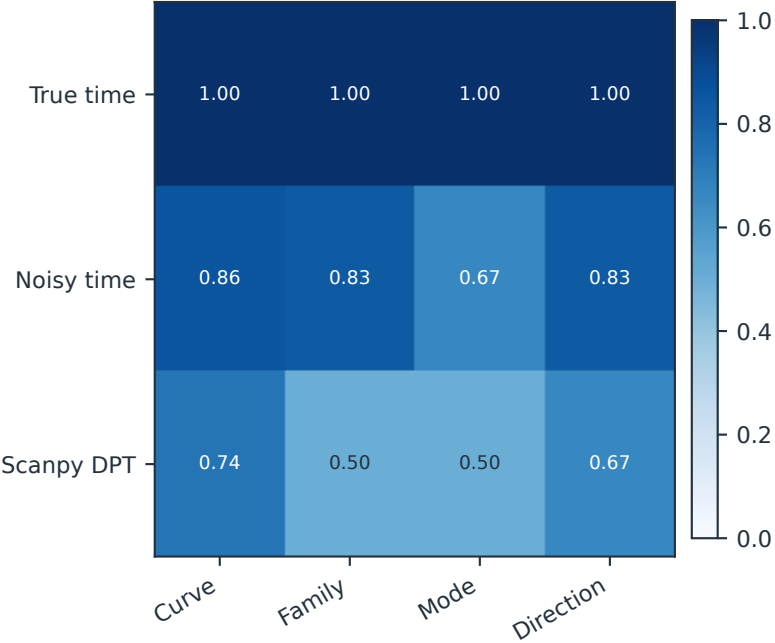
A. ZSCAPE exact leave-one-embryo stability



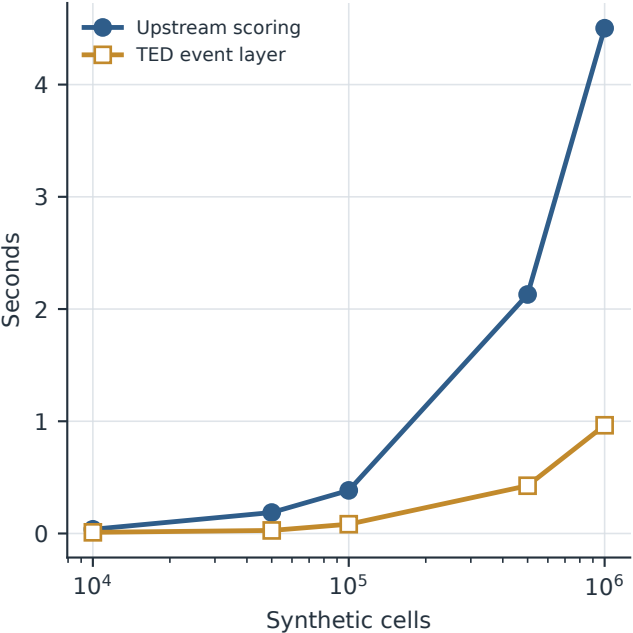
B. Four-dataset frozen-gate audit



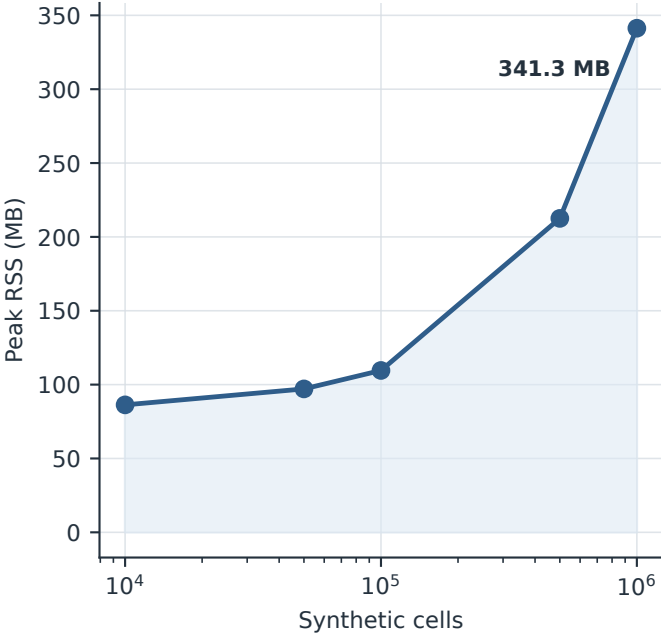
C. PyFgsea sensitivity to trajectory input



D. Scaling time by component



E. Peak resident memory



F. Executable native package adapters

Package	Version	Runtime
tradeSeq	1.24.0	9.42 s
GSVA	2.4.4	8.13 s combined
AUCell	1.32.0	8.13 s combined
POT	0.9.6.post1	1.41 s

All exited 0 and recovered the prespecified native toy signal.